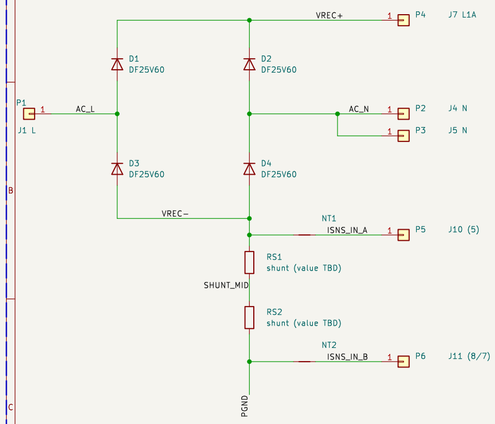
LeafOBC-WafflePlate: first-pass schematic of the 2012 Nissan Leaf OBC power module ("waffle plate")

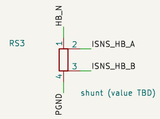
DRAFT. Sections A and D are meter-verified and drawn with wires; the rest uses net labels. Topology is from the community draft schematic plus identified part numbers. Connections are NOT yet verified against the module. Hidden fields per part: Verified, Draft_ref, Location_px (pixel position in images/stitched/waffle-plate-top.jpg).

A. Input bridge and shunts (upper left)



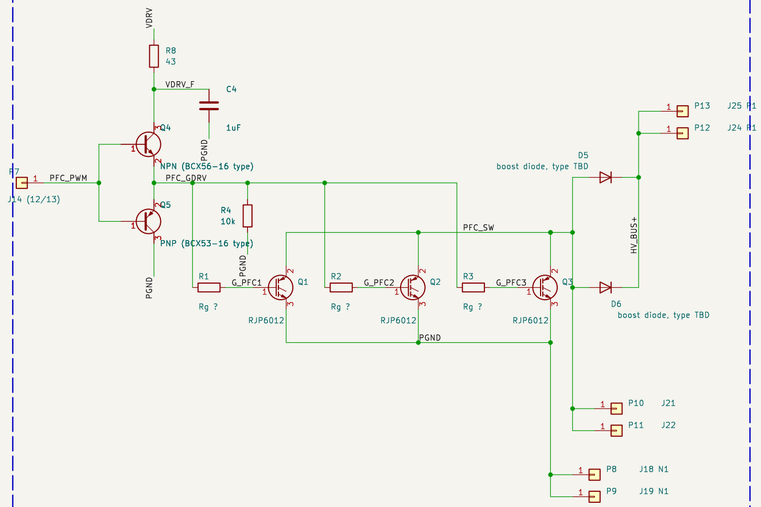
RS1 + RS2: one current shunt in two parts.

Kelvin sense taken from the outer terminals (NT1, NT2).



RS3: full-bridge return (section C)

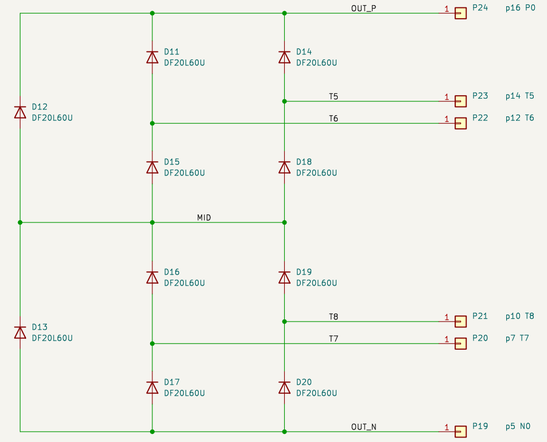
B. PFC boost stage (mid left)



Q1–Q3 in parallel, each with its own gate resistor. D5, D6 in parallel.

Gate driver (Q4, Q5, R4, R8, C4) meter-verified. IGBTs, gate resistors and boost diodes believed correct, not yet meter-verified.

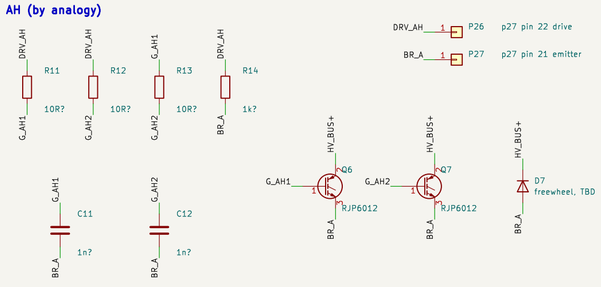
D. Output rectifier: two series-stacked bridges (upper right)



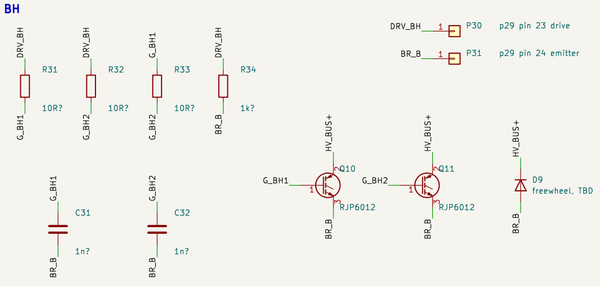
Verified with meter: D11–D20 and pads.

C. Primary full bridge: 4 switches, each 2 IGBTs in parallel + gate network (lower right)

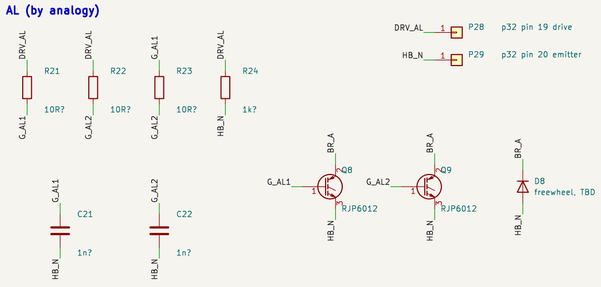
AH (by analogy)



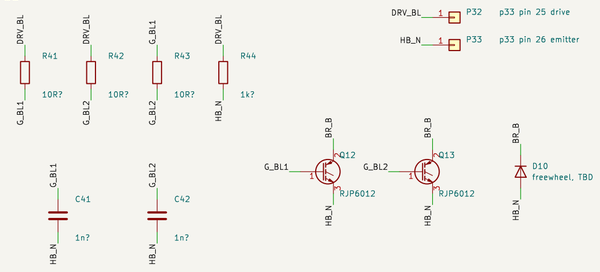
BH



AL (by analogy)

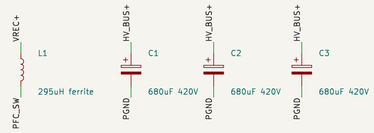


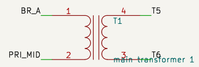
BL

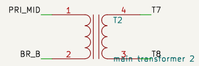


Values with ? are reference values (Ukrainian schematic and replica), unconfirmed. AL (Q8, Q9) failed on the reference module. See docs/full-bridge-gate-drive.md

E. Off-module parts (charger PCB)

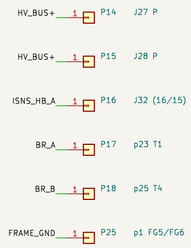






Primaries shown in series: verify.

F. Remaining module pads (draft numbering, unverified)



Pads for A, B and D are drawn next to their circuit.

Unassigned draft pads: J8 (6, IC312_PFC), J12/J13 (11), J15 (10/9), J30/J31 (N–N1), p31 (17/18, thermistor TH202, not yet drawn)

CERN-OHL-S-2.0

Sections A and D meter-verified. Rest unverified.

LeafOBC-WafflePlate

Sheet: /

File: waffle-plate.kicad_sch

Title: Leaf OBC waffle plate: original module

Size: A2	Date:	**Rev: 0.1-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1